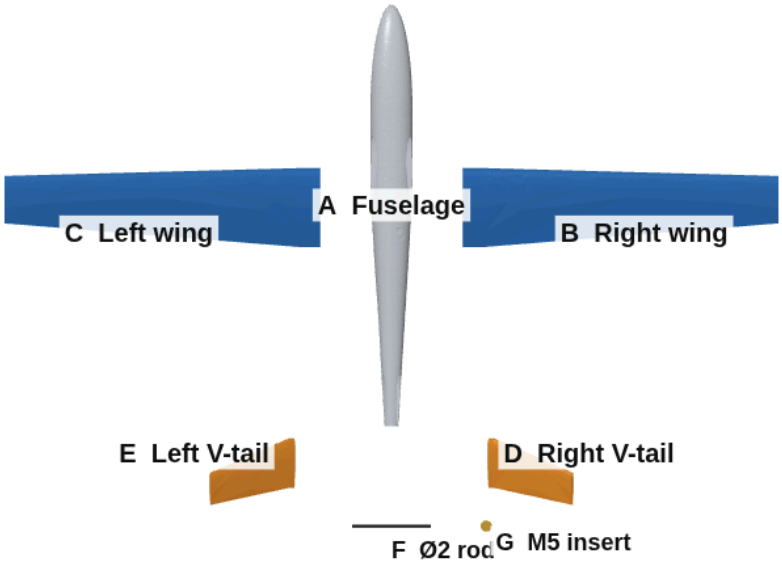
Stravion Aerospace UAV: 1:6 Wind-Tunnel Model

Assembly guide and bill of materials. Covers the printed test model only (no propulsion, avionics or control surfaces). Span 400 mm, length 277 mm, finished mass ≈ 189 g in PLA.



1. Bill of materials

Printed parts (from `parts/`)

ID	Part	File	Qty	Size (mm)	Mass, PLA 100 %
A	Fuselage	`fuselage.stl`	1	267 × 27 × 27	106 g
B	Right wing	`wing_right.stl`	1	200 × 50 × 11	35 g
C	Left wing	`wing_left.stl`	1	200 × 50 × 11	35 g
D	Right V-tail panel	`vtail_right.stl`	1	42 × 55 × 38	4 g
E	Left V-tail panel	`vtail_left.stl`	1	42 × 55 × 38	4 g
Total printed (excl. brim/supports)					**184 g**

Hardware

ID	Item	Qty	Notes
F	Carbon-fibre rod, Ø 2.0 mm (pultruded)	50 mm	Wing joiner. Buy a 2 × 500 mm rod and cut it with a hobby knife or cutting disc. A Ø 2 mm steel rod or drill blank also works. The holes are Ø 2.15 mm.
G	Brass heat-set insert, M5 × 9.5 mm (e.g. ruthex RX-M5x9.5)	1 (buy a pack)	Needs a Ø 6.4 mm hole, which is what the model has. Practise on a scrap print first.
H	M5 stainless bolt or threaded rod + M5 jam nut	1 + 1	Length to suit your balance/sting adapter. This is the only connection to the balance.

Consumables

Item	Amount	Notes
Filament: **PLA or PLA+** (recommended)	≈ 230 g incl. brim/supports	A 1 kg spool makes about 4 complete models (handy for spares/variants). PLA is the stiffest common filament, which keeps wing bending lowest. PETG is tougher but about 35 % more flexible. Avoid carbon-filled filaments unless you sand them smooth.
2-part epoxy, **30-minute** type	≈ 5 g	For the wing joint and rod. 30-minute epoxy gives you time to align. 5-minute works but is rushed.
Medium cyanoacrylate (CA) glue	a few drops	Optional, for the V-tail panels (epoxy is fine too).
Isopropyl alcohol + paper towels	—	Degrease parts and wipe away epoxy squeeze-out.
Wet/dry sandpaper 220 / 400 / 600 / 1000 + small sanding block	1 sheet each	The surface finish matters at these Reynolds numbers.
Sandable filler primer (spray) + spot putty	optional	Fills layer lines and seams. Keep the total coating under ~0.1 mm.
Low-tack masking tape	1 roll	Holding parts while glue cures, masking, and guiding trip-strip placement.
Boundary-layer trip: zig-zag turbulator tape, ≈ 0.2–0.3 mm thick, 3–5 mm wide	≈ 1 m	Alternative: #120 carborundum grit on thin double-sided tape. Check that it works with tufts or oil flow.

Flow-vis supplies (fine wool tufts, oil-flow mix), matte paint	optional	Matte black for oil flow, white for tufts.

Tools

FDM printer, bed ≥ 220 × 220 mm (fuselage goes diagonally), Z ≥ 210 mm (wings print standing)	Digital calipers
Soldering iron with an M5 heat-set tip (or a conical tip)	Drill bits Ø 2.2 mm and Ø 6.4 mm, turned *by hand* or in a pin vise to clean holes
Hobby knife, needle files, flush cutters	Flat reference surface (glass/stone tile), small square, digital angle gauge or protractor
Mixing cups and sticks, nitrile gloves	Foam blocks or a soft towel to cradle the fuselage

2. Printing

Part	Orientation	Supports	Notes
Wings B, C	**Standing on the root face** (flat face down), both on the plate together	None	Every layer is a true airfoil section, giving the best leading edge, no support marks, and an accurate vertical rod hole. Use an 8–10 mm brim. Slow the last 50 mm (a tall, slim part).
V-tail D, E	Standing on the flat root-tab face	None	5 mm brim. 0.08–0.12 mm layers (thin part).
Fuselage A	Lying on its side, diagonally on the bed (wing slot vertical)	Tree supports, **from build plate only**	Block supports inside the slot and the V-tail pockets. Afterwards, ream the Ø 6.4 mm belly hole by hand if it printed slightly oval. 5 mm brim.

Common settings: 0.4 mm nozzle · 0.12 mm layers · 100 % infill (or ≥ 6 walls + 40 % gyroid for the fuselage) · 6 top/bottom layers · **seam position: rear / aligned on the trailing edge** (keeps seams off the airfoil surfaces) · outer walls 30–40 mm/s · full part cooling.

3. Before you glue: inspect & dry-fit

☐ Remove brims and supports. Clean the trailing edges with a sanding block (don't round them; they are 0.6 mm thick by design).

☐ Check key dimensions with calipers. Wing: root chord 50.0 mm, tip chord 30.0 mm, length 200 mm. V-tail: root 31.3 mm, tip 20.4 mm.

☐ Turn a Ø 2.2 mm drill by hand through both wing rod holes. Test the rod: snug, not forced.

☐ Slide each wing stub into the fuselage slot and each V-tail tab into its pocket. The target is light hand pressure. If tight, sand the *part* (stub or tab) with 400 grit, not the slot. If there is visible play (> 0.3 mm), plan on using epoxy, which fills gaps.

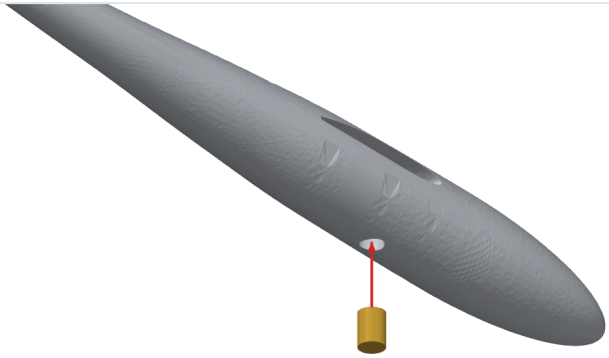
☐ Rough-sand the external surfaces now (220 → 400). Leave the stubs, tabs and slot interiors unsanded except for fit.

4. Assembly

1 Heat-set insert (do this first)

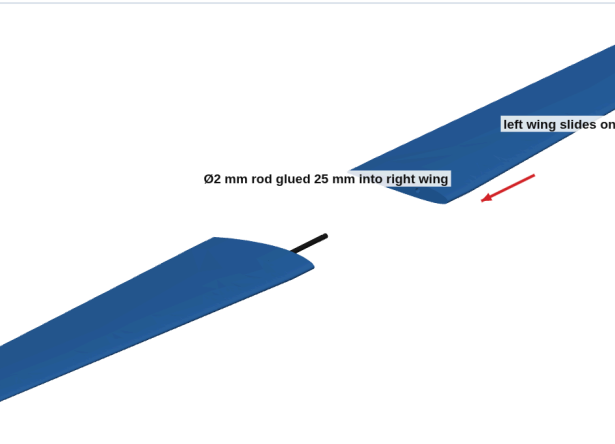
- Cradle the fuselage upside-down on a towel or foam, belly hole facing up and vertical.
- Iron at ~210 °C for PLA (~240 °C for PETG). Set insert G on the hole and let it sink under light pressure. **Keep it perpendicular.**
- Stop when it is flush with the surface. Don't push deeper: the wing slot is about 6 mm above the hole.
- While it's still warm, thread in an M5 bolt to square it up. Remove the iron and let it cool for 2 minutes.
- Look into the wing slot. If any plastic bulged in, trim it with a knife.

> Doing this before gluing means the heat can't disturb any glued joints, and the fuselage is still easy to hold.

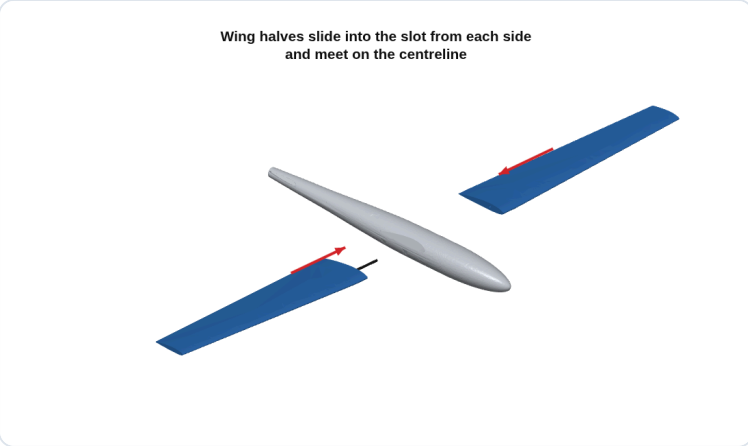


2 Joiner rod into the right wing

- Cut 50 mm of the Ø 2 mm rod and lightly scuff it with 220 grit.
- Mix a little 30-minute epoxy. Coat the rod for 25 mm and push it fully into the **right** wing's hole (it bottoms at 25.5 mm).
- Wipe off the excess. Stand the wing root-up so the rod stays straight while it cures.
- 25 mm of rod should stick out. Don't glue the left wing yet.



3 Wings into the fuselage

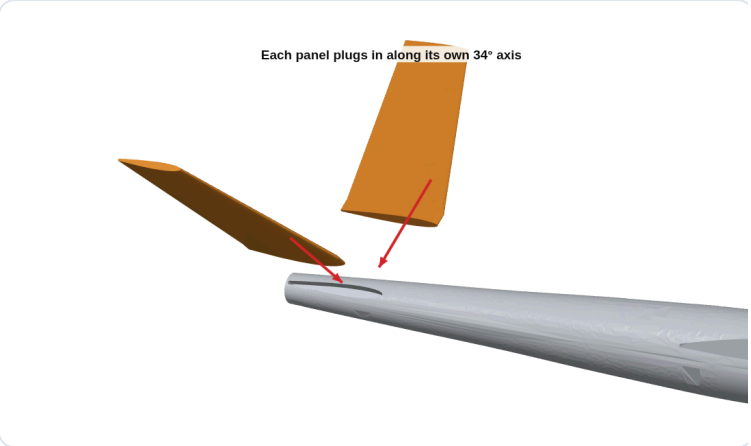


Dry run first: slide the right wing into the slot from the right until the rod reaches the centreline. Then slide the left wing in from the left, over the rod, until the two root faces meet. Check:

- The root faces touch with no gap at the centreline (look through the slot).
- Level the fuselage (roll) in a cradle on a flat surface. Measure the height of each wingtip trailing edge from the table: **left = right within 0.5 mm**. Each tip sits about 6.5 mm above the root because of the 2° dihedral.
- Looking from the front, both tips show the same twist (2° washout).

Glue: take it apart and apply a thin film of epoxy to both wing stubs, the root faces and the left-wing rod hole. Re-insert, press the roots together, and wipe the squeeze-out at the fuselage junction with IPA. Tape across the joint. Prop the tips to equal height and let it cure (full strength in about 24 h before tunnel loads).

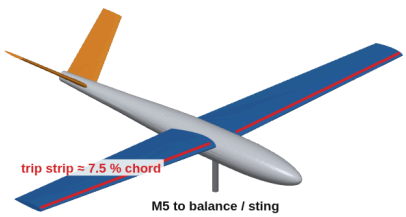
4 V-tail panels



- Dry-fit each panel. The tab slides straight in along the panel's own 34° axis until it bottoms out. The panel root should sit flush with the fuselage surface.
- Check the angle: **34° above horizontal** (112° between the panels). Use a digital angle gauge on the panel with the fuselage levelled.
- Check symmetry: each tip is about 37 mm above the fuselage axis and 55 mm out from the centreline. Left and right should match within 0.5 mm.
- Put a drop of medium CA (or epoxy) in the pocket, insert, and hold or tape until set. Do one side at a time.

5 Finish, trips & mounting

- **Seams:** fill only actual gaps at the wing and V-tail junctions with spot putty, then wet-sand 400 → 600 → 1000. Optionally apply 1–2 light coats of filler primer and sand again. Plug the M5 insert with a bolt while spraying.
- **Trip strips:** apply at **≈ 7.5 % of local chord** on the upper and lower wing surfaces. That's 3.8 mm from the leading edge at the root and 2.3 mm at the tip. Use a strip of masking tape as a straight-edge guide. The V-tail is optional, at the same 7.5 %.
- **Mark the moment reference point:** x = 95.83 mm from the nose (25 % MAC), on the insert axis, 5.0 mm above the fuselage axis. That puts it **18.3 mm above the belly surface at the insert**. Use this distance to transfer balance moments.
- **Mount:** thread the M5 bolt/rod into the insert and lock it with a jam nut against your adapter, not against the plastic. Snug only: ≤ ~1.5 N·m. Align zero roll (wings level), zero yaw, and set pitch on the fuselage axis.



5. Operating limits & checks

Estimated wingtip deflection (solid print)	20 m/s, C_L 0.6	20 m/s, C_L 1.1	30 m/s, C_L 0.6	30 m/s, C_L 1.1
PLA	1.0 mm	1.8 mm	2.2 mm	3.9 mm
PETG	1.5 mm	2.8 mm	3.4 mm	6.2 mm

Recommendation: PLA, at ≤ 25 m/s for high-α sweeps. At 30 m/s near $C_{L,max}$, the tips bend about 2 % of the semi-span. That adds roughly 1° of effective dihedral and slightly affects roll and yaw data. If you need higher dynamic pressure, I can add a full-length carbon spar channel to the wings. PLA softens around 55 °C, so keep the model out of hot cars and sunlit windows.

☐ Before each test session: wiggle-test the wings and V-tail, check the trip strips are intact, and re-check that the tip heights are symmetric.

☐ Record the as-built mass and the measured incidence and dihedral with your data.

☐ Run a tare (wind-off) at each attitude: the model weighs ≈ 189 g, and its CG is not at the reference point.

6. Quick shopping checklist

- ☐ PLA/PLA+ filament, 1 kg spool
- ☐ Ø 2 mm pultruded carbon rod (500 mm)
- ☐ M5 × 9.5 mm brass heat-set inserts (pack)
- ☐ M5 stainless bolt/threaded rod + jam nut (to suit your balance)
- ☐ 30-minute epoxy · medium CA · isopropyl alcohol
- ☐ Wet/dry sandpaper 220/400/600/1000 · sanding block · spot putty · filler primer (optional)
- ☐ Zig-zag turbulator tape 0.2–0.3 mm · low-tack masking tape
- ☐ Heat-set tip for the soldering iron · Ø 2.2 and Ø 6.4 mm drill bits · calipers · angle gauge